Ashmit Dutta

612-323-1292 | ashmitd2@illinois.edu | ashmitdutta.xyz | linkedin.com/ashmitdutta | github.com/aedutta

Education

University of Illinois Urbana-Champaign Urbana, IL
BSc, Computer Engineering *Aug 2023 – May 2027*

- **Publications:** *Teaching Reasoning to LLMs* (ICML 2025 Workshop); *AI for 3D Shape Acquisition* (ISC 2023)

Experience

Assured and Trustworthy AI Research Lab, UIUC Urbana, IL
Undergraduate Research Assistant (Prof. Huan Zhang) *Dec 2025 – Present*

- Benchmarking formal verification performance on large-scale deep learning models.

John Deere Champaign, IL
Software Engineer, Part-Time Student *Jan 2025 – Jan 2026*

- Modernized exporting platform for 500+ users, migrating legacy stack to React, Java Spring Boot, and AWS.
- Designed and implemented the core API, integrating data across PostgreSQL and DB2 for millions of records.
- Integrated AWS CloudWatch for real-time observability, reducing incident resolution time (MTTR) by 40%.
- Engineered automated CI/CD pipelines to cut deployment time 50% and secured sensitive export data on AWS S3.

The University of Chicago Data Science Institute Chicago, IL
Machine Learning Research Intern (Summer REU) *Jun 2024 – Aug 2024*

- Built a multi-modal GNN in PyTorch Geometric for Fermilab's Exa.TrkX neutrino project.
- Expanded architecture with message-passing layers, improving model accuracy to 98.3% (3 graph layers).
- Processed 20,000+ neutrino interactions using H5Py on UChicago's HPC cluster (Slurm).

Projects

RISC-V Operating System | *C, Assembly, QEMU* Mar 2025 – Present

- Designed a Unix-like OS from scratch in C for RISC-V with paging, multitasking, file I/O, and shell support.
- Implemented drivers (UART, VIRTIO, RTC, NIC), system calls, ELF program loading, and Sv39 virtual memory.
- Developed fork/exec support, pipes, and a trap-based preemptive scheduler; debugged using GDB.
- Currently developing a bare-metal hypervisor to run multiple OSes concurrently.

Ultra-Low Latency Trading Engine | *C++20, AWS, Linux, Network Performance* Dec 2025 – Present

- Engineered a cloud-native HFT engine for Coinbase, achieving 36ns median internal logic latency on AWS c7i.large.
- Implemented thread-per-core, lock-free design using hugepage-backed SPSC ring buffers and CPU pinning.
- Developed execution gateway with Boost.Beast/OpenSSL using TCP_NODELAY and JWT caching.

Autonomous Drone Racing Stack | *ROS 2, Python, C++, PyTorch* Oct 2025 – Dec 2025

- Developed 500 Hz ROS 2 autonomy stack for Crazyflie 2.1 integrating Geometric Controller and CasADi MinSnap.
- Developed Python hardware interface bridging planners to embedded firmware via CRTP radio.
- Finetuned YOLOv8-seg gate segmentation model with PyTorch and multi-Bayesian optimizer on NeRF data.

Technical Skills

Languages: C/C++, Python, Java, JavaScript, SystemVerilog, SQL, Assembly (RISC-V)
Frameworks: React.js, Spring Boot, PyTorch, TensorFlow, ROS 2, Flask, Pandas, NumPy
Tools: Docker, AWS, Linux/Unix, GDB, Git, LaTeX, Slurm, QEMU, Wireshark

Leadership, Publications & Awards

Founder, Online Physics Olympiad: Led 20k+ community, raised $10k+ (Citadel/Jane Street).

Executive Board Member (Treasurer), IEEE Chapter, UIUC

Course Staff: Course Assistant for CS 450 (Numerical Analysis).

Awards: 2nd Place Bradley Mottier Innovation Challenge ($2000), USA Computing Olympiad Gold.